

by it. In this mode, you need to know the password for the network in order to connect to the WAP. In the second method, known as the monitor mode, you do not have to connect to the WAP to intercept the data; yet you can monitor all the traffic.

However, these modes are not supported by all the wireless network cards. For example, Intel's 802.11g cards do not support the 'promiscuous mode'. The monitor mode also needs to be supported by the card. The advantage of the monitor system (from a cracker's perspective) is that it does not leave any trace on the WAP—no logs, no transfer of packets to the WAP or directly from the WAP.

Wireless sniffing: a case study

Sniffing wireless networks is more complicated than sniffing wired networks. This is mainly because of the various encryption protocols used. If you want to sniff a network with Wired Equivalent Privacy (WEP) security then it is fairly easy. In fact, it has been proved many times that WEP can be easily cracked (as will be shown later in the article). Sniffing/cracking networks with Wireless Protected Access (WPA) security, however, is not so easy.

The difference in WPA and WEP is that WEP applies a static method to use pre-shared keys for encryption. It uses the same key to encrypt all the data. This means a large number of packet transfers with the same key, which makes cracking easy. Second, one has to manually update all the client machines when a WEP key is changed on the network. This is not practical for large installs. WPA, on the other hand, uses the pre-shared keys to derive a temporary key, using which all the traffic is encrypted. So, WPA generates a unique key for each client and access point link. Moreover, the pre-shared key is very rarely used, making it difficult for sniffers to crack the key. I would like to make one point clear here—one can crack WPA passwords if they are too simple. This is not a flaw in WPA, but in the network manager who sets the weak password.

We will now see how to sniff a wireless network with WEP security and use the sniffed packets to crack the password.

For this study, I will be using two laptops. One running a Live CD of BackTrack Linux 3 and the other running Windows XP. The Windows laptop has access to the WAP. The user knows the key. He is using the Internet on his laptop. I (the cracker) am using the laptop with BackTrack Linux. There are many popular wireless sniffing and key sniffing tools available for Linux like Air Snort, Air Crack, WireShark, etc. I decided to go with Air Crack. (For an extensive list of all the tools, please visit, backtrack.offensive-security.com/index.php?title=Tools#Radio_Network_Analysis).

Remember, not all cards support monitor mode, which is what is being used here to crack the password. I am not going into the details of how to install Air Crack (or any other tool) in this article. I assume that you already have the software. In order to carry out attacks on wireless networks efficiently, you'll almost certainly need to patch your wireless drivers to support packet injection—the

```
bt ~ # iwconfig
lo          no wireless extensions.

eth0       no wireless extensions.

eth1       unassociated  ESSID:off/any
          Mode:Managed  Channel:6    Access Point: Not-Associated
          Bit Rate:0 kb/s Tx-Power:20 dBm Sensitivity=8/0
          Retry limit:7   RTS thr:off   Fragment thr:off
          Encryption key:off
          Power Management:off
          Link Quality:0 Signal level:0 Noise level:0
          Rx invalid mid:0 Rx invalid crypt:0 Rx invalid frag:0
          Tx excessive retries:0 Invalid misc:0 Missed beacon:0

bt ~ #
```

Figure 1: A typical iwconfig output

```
bt ~ # airodump-ng start eth1

Interface   Chipset      Driver
eth1        Centrino b/g ipw2200 (monitor mode enabled)

bt ~ #
```

Figure 2: Starting the monitor mode

patches as well as details of how to do this can be found at www.aircrack-ng.org/doku.php?id=install_drivers. BackTrack Linux comes with pre-patched drivers and is a very good distribution for hacking purposes. The driver being used in this experiment is 'MadWiFi'.

Now you can check if your card supports monitor mode by issuing the following command as the root user (from here on, all the commands are issued as the root):

```
iwconfig
```

This will give you the name of your wireless network card (Figure 1). Once you get that, issue the following:

```
airmon-ng stop eth1
```

You can replace 'eth1' with the name of your wireless network card device.

Then execute the following command to make eth1 work in 'monitor' mode (Figure 2):

```
airmon-ng start eth1
```

Now scan for wireless access points by issuing the following command:

```
airodump-ng eth1
```

As you can see in Figure 3, this will show you any networks detected, the MAC addresses of the access points (BSSID), the MACs of any computers that are connected to them (STATION), and the Wi-Fi channels they are operating on. If the access point is broadcasting its name (ESSID), this will also be shown.

Once you have got this information, you can try and crack the key. Note the channel of the WEP encrypted network in Figure 3—it is 6. Quit *airodump* by pressing Ctrl+C and then issue the following: